

MATH 3260 - Chapter 10 Worksheets

Numerical Methods

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 10.1 - Gaussian Elimination with Partial Pivoting

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.
Use the system below for Problems 3-5:

$$0.002x_1 + 2x_2 - 2x_3 = 6.002,$$

$$3x_1 - 4x_2 - 4x_3 = -1,$$

$$x_1 + 4x_2 - 4x_3 = 13.$$

Its exact solution is $(1, 2, -1)^T$.

1. Round 0.009995, 18.245, and -763.55 to four significant digits using round-half-to-even.

2. Explain catastrophic cancellation using

$$1.234567 - 1.234566 = 0.000001.$$

Why can the relative accuracy of the difference be much worse than the relative accuracy of the two inputs?

3. Reproduce the nonpivoted elimination using four-significant-digit arithmetic. Record the multipliers and the approximate solution.

4. Repeat Problem 3 with partial pivoting. Compare the pivot choices and multiplier magnitudes.

5. Compute the residual and forward error for both approximations. Explain why a small residual does not automatically guarantee a small forward error.

6. Prove the perturbation bound

$$\frac{\|\Delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(A) \frac{\|\Delta \mathbf{b}\|}{\|\mathbf{b}\|}, \quad \kappa(A) = \|A\| \|A^{-1}\|.$$

7. Explain why partial pivoting improves algorithmic stability but cannot repair an ill-conditioned problem.

Section 10.2 - Iterative Methods for Linear Systems

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. Use

$$\begin{aligned}10x_1 - x_2 + 2x_3 &= 6, \\ -x_1 + 11x_2 - x_3 &= 22, \\ 2x_1 - x_2 + 10x_3 &= -10\end{aligned}$$

for Problems 1-4. The exact solution is $(1, 2, -1)^T$.

1. Derive the Jacobi and Gauss-Seidel component update equations.
2. Starting from $\mathbf{x}^{(0)} = \mathbf{0}$, compute three iterations of each method by hand.
3. Compute the iteration matrices B_J and B_{GS} and their spectral radii.
4. Prove convergence from strict diagonal dominance and compare the observed convergence speeds of Jacobi and Gauss-Seidel.
5. Consider

$$x + 4y = 5, \quad 3x + y = 4.$$

Demonstrate divergence of Jacobi in the displayed order from $(0, 0)$, then reorder the equations and show why the reordered system converges.

6. Design a stopping rule that checks both iterate change and residual. Explain why using both is safer than using only one diagnostic.

Method choice: direct methods are usually attractive for small/medium dense systems or repeated solves after LU; iterative methods can be attractive for very large sparse systems when convergence is favorable.

Section 10.3 - Power Method for Dominant Eigenvalues

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation. Use

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \quad \mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

for Problems 1-4.

1. Find the exact eigenpairs of A .

2. Perform six scaled power iterations from $\mathbf{x}^{(0)}$.

3. At each step, compute the Rayleigh quotient and eigen-residual

$$\|A\mathbf{x}^{(k)} - \rho_k \mathbf{x}^{(k)}\|.$$

4. Prove the power-method convergence formula for a diagonalizable matrix with a unique dominant eigenvalue.

5. Compare two diagonalizable matrices whose dominant-to-subdominant magnitude ratios satisfy

$$\left| \frac{\lambda_2}{\lambda_1} \right| = 0.2 \quad \text{and} \quad \left| \frac{\lambda_2}{\lambda_1} \right| = 0.9.$$

Which should converge faster, and why?

6. Give an initial vector for which the power method misses the dominant eigenvalue. Explain the missing-component mechanism.

7. Compare the normalization scale factor with the Rayleigh quotient as eigenvalue estimates. When is the Rayleigh quotient especially attractive?

Section 10.4 - Applications of Numerical Methods

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Hiking-elevation data for Problems 1-2. At hours $x = 0, 1, 2, 3, 4, 5$, the recorded elevation gains are

$$120, 182, 255, 329, 388, 415.$$

The quadratic model is $p(x) = a_0 + a_1x + a_2x^2$.

Probability model for Problems 3-4.

$$p_1 = \frac{1}{2}p_2, \quad p_2 = \frac{1}{3}(p_1 + p_3 + p_4), \quad p_3 = \frac{1}{2}(p_2 + 1), \quad p_4 = \frac{1}{3}(p_2 + 1 + 0).$$

The exact solution is $(1/4, 1/2, 3/4, 1/2)$.

Trail-songbird model for Problem 5.

$$L = \begin{bmatrix} 0 & 1.1 & 1.6 & 0.5 \\ 0.72 & 0 & 0 & 0 \\ 0 & 0.65 & 0 & 0 \\ 0 & 0 & 0.55 & 0 \end{bmatrix}.$$

1. Derive the quadratic normal equations from the sum of squared residuals for the hiking-elevation data.
2. Solve the normal equations with partial pivoting and predict the elevation gain at $x = 6$. Comment on the limitations of extrapolating beyond the observed data range.
3. Rewrite the four probability equations as a linear system $A\mathbf{p} = \mathbf{b}$.
4. Starting from $\mathbf{p}^{(0)} = \mathbf{0}$, perform eight Gauss-Seidel iterations for the probability system and compare with the exact solution.
5. Use the power method to estimate the dominant growth factor of the trail-songbird Leslie matrix and the corresponding stable age proportions.
6. For each of the three applications above, list at least three model limitations or assumptions that should be checked before interpreting the numerical result literally.